Sesi 7

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Open-Economy Macroeconomics: Basic Concepts

Pengantar Ekonomi 2

Sumber: Mankiw (Ch 32)

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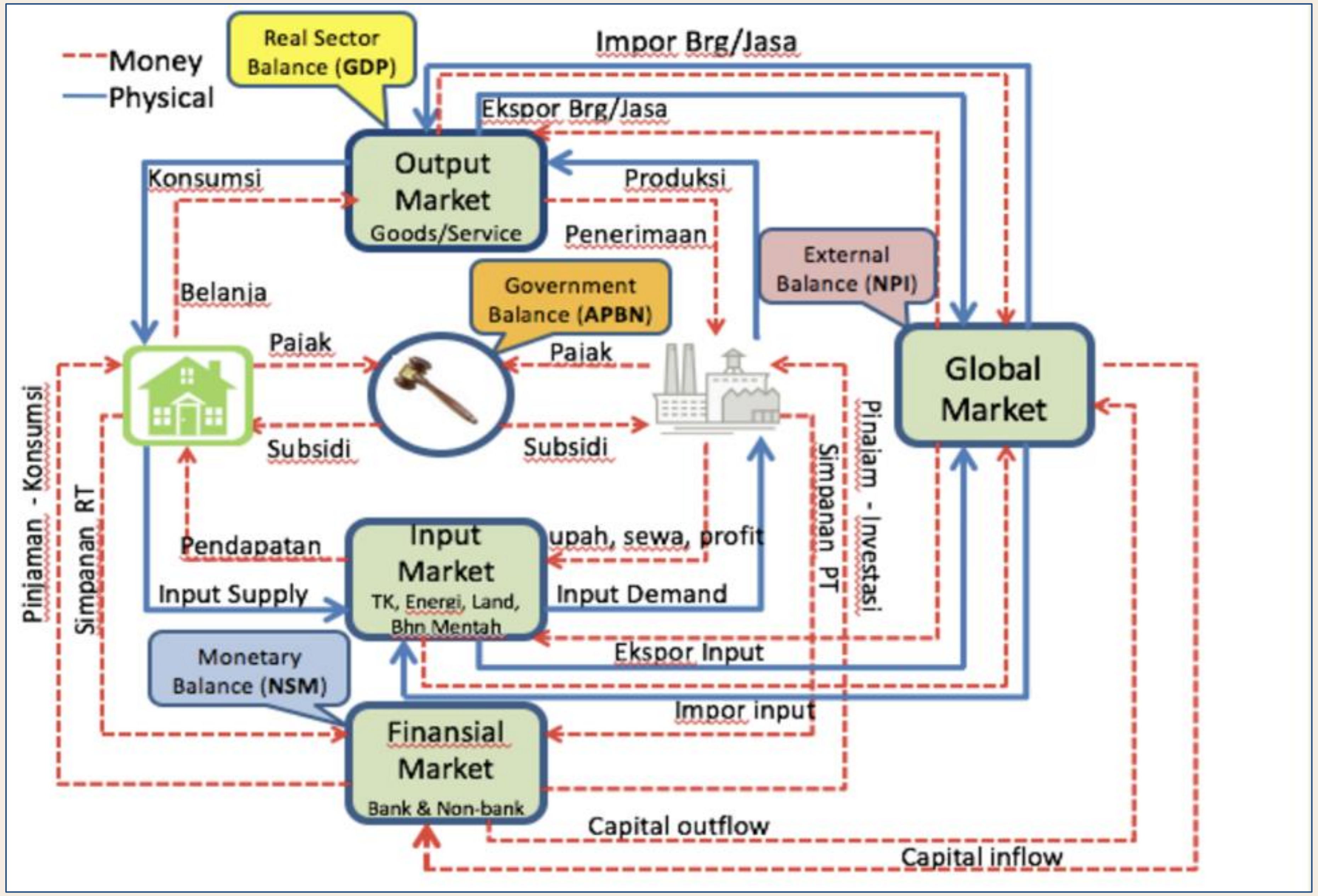
Introduction

- Closed economy
  - Economy that does not interact with other economies in the world
- Open economy
  - Economy that interacts freely with other economies around the world
  - **Buys and sells goods and services** in world product markets
  - **Buys and sells capital assets** such as stocks and bonds in world financial markets



Hubungan Antar Variabel Makroekonomi
Source: Zakir Machmud

International Flow of Good and Capital

The Flow of Goods: Export, Import, and Net Export

Recall that GDP is:
$Y = C + I + G + NX$

Export
Goods and services that are produced domestically and sold abroad

−

Imports
Goods and services that are produced abroad and sold domestically

=

Net Export (NX)
Value of exports – value of imports

Trade surplus, *NX* > 0
- The country sells more goods and services abroad than it buys from other countries

Trade deficit, *NX* < 0
- The country sells fewer goods and services abroad than it buys from other countries

Balanced trade:
Exports = Imports

Trade Balance

*Factors that influence import, export, and net exports include **consumer taste** for goods and services**, the prices of goods** at home and abroad**, the exchange rate** people use to buy foreign currency, **the incomes** at home and abroad, **the cost of transport**, and **government policies**.*

Active Learning 1: Variables that affect ***NX***

What do you think would happen to U.S. net exports if:

A. Canada experiences a recession (falling incomes, rising unemployment)

B. U.S. consumers decide to be patriotic and buy more products "Made in the U.S.A."

C. Prices of goods produced in Mexico rise faster than prices of goods produced in the U.S.

Answers:

A. Canada experiences a recession (falling incomes, rising unemployment)
- U.S. net exports would fall
- due to a fall in Canadian consumers' purchases of U.S. exports

B. U.S. consumers decide to be patriotic and buy more products "Made in the U.S.A."
- U.S. net exports would rise
- due to a fall in imports

C. Prices of goods produced in Mexico rise faster than prices of goods produced in the U.S
- This makes U.S. goods more attractive relative to Mexico's goods.
- Exports to Mexico increase, imports from Mexico decrease, so, U.S. net exports increase.

The Flow of Financial Resources

Net Capital Outflow (NCO) = Purchase of foreign asset by domestic residents - Purchase of domestic assets by foreigners.

Purchase of foreign assets by domestic residents − Purchase of domestic assets by foreigners = Net capital outflow, ***NCO*** - (net foreign investment)

- FDI
  - Contoh: orang asing membuka pabrik di Indonesia.
- FPI
  - Contoh: orang asing membeli saham di Indonesia

When ***NCO*** > 0, **"capital outflow"**

- Domestic purchases of foreign assets exceed foreign purchases of domestic assets

When ***NCO*** < 0, **"capital inflow"**

- Foreign purchases of domestic assets exceed domestic purchases of foreign assets

Factors that influences NCO are: ***real interest rate of foreign asset**, **real interest rate of domestic asset**, **economic and political risk of having assets abroad**, and **government policies surrounding foreign ownership of domestic asset.***

The Equality of NX and NCO

An accounting identity: ***NCO = NX***

Every transaction that affects *NX* also affects *NCO* by the same amount (and vice versa)

Ketika *foreigner* membeli barang dari Indonesia,
- Maka ekspor dan nilai *net export* (NX) meningkat.
- *Foreigner* tersebut membayar dengan mata uang atau aset, sehingga Indonesia memperoleh sejumlah *foreigner asset*, yang menyebabkan Net Capital Outflow (NCO) meningkat.
- X↑z, NX↑, *foreign asset* ↑, NCO↑

Ketika orang Indonesia membeli barang dari luar negeri
- Impor meningkat sehingga *net export* (NX) menurun.
- Pembayaran yang dilakukan menggunakan mata uang atau aset domestik menyebabkan pihak luar negeri memperoleh aset Indonesia, sehingga menurunkan NCO Indonesia
  M ↑ → NX ↓ → *foreigner owns more Indonesian assets* ↑ → NCO ↓

The Equality of NX and NCO

Persamaan identitas: ***Net Export = Net Capital Outflow***

Recall that:

- Open economy: $Y = C + I + G + NX$
- National saving: $S = Y - C - G$
  - $Y - C - G = I + NX$
  - $S = I + NX$
- $NX = NCO$
  - $S = I + NCO$
  - Saving = Domestic investment + Net capital outflow

- $S = I + NX$
- $S - NX = I$
- $S - NCO = I$
- $S + NCI = I$

Trade Surplus and *NCO*

A country running a trade surplus, $NX > 0$

- Negara tersebut mengekspor lebih banyak barang dan mengakumulasi aset luar negeri
- Artinya modal mengalir ke luar negeri.
- Sehingga **NCO > 0**

Trade Deficit and *NCO*

- A country is running a trade deficit, $NX < 0$
  - Artinya negara tersebut **mengimpor lebih banyak barang daripada yang diekspor**.
  - Bagaimana hal itu dibiayai?
  - Defisit tersebut harus dibiayai oleh **aliran masuk modal (*capital inflows*)**—artinya pihak asing membeli aset domestik.
  - Oleh karena itu, **NCO < 0**.

Implications

1. Ketika sebuah negara **surplus dalam NX**, maka **NCO pasti positif**. Artinya, jumlah pembelian aset luar negeri lebih besar dari pada orang asing beli aset domestik.

1. Ketika sebuah negara **defisit dalam NX,** maka **NCO pasti negatif**. Artinya, lebih banyak orang asing membeli aset domestik dibandingkan orang domestik membeli aset luar negeri.

Summary

TABLE 1

International Flows of Goods and Capital: Summary
This table shows the three possible outcomes for an open economy.

Trade Deficit	Balanced Trade	Trade Surplus
Exports < Imports	Exports = Imports	Exports > Imports
Net Exports < 0	Net Exports = 0	Net Exports > 0
$Y < C + I + G$	$Y = C + I + G$	$Y > C + I + G$
Saving < Investment	Saving = Investment	Saving > Investment
Net Capital Outflow < 0	Net Capital Outflow = 0	Net Capital Outflow > 0

EXAMPLE 1: Exporting to Germany

Kiara is a web designer living in Illinois. She creates and sells a website to Gabrielle who is living in Germany. Gabrielle pays Kiara 5,000 euros for the website. What is the effect on the U.S. net exports and net capital outflows if:

A. Kiara keeps the 5,000 euros at home
B. Kiara buys 5,000 euros worth of stocks in a German company
C. Kiara spends the 5,000 euros on shoes made in Germany
D. Kiara exchanges the 5,000 euros into U.S. dollars

Kiara (U.S.) sells a website to Gabrielle (Germany) for 5,000.

- ***NX*** = Exports – Imports
- ***NCO*** = Purchases of foreign assets by domestic resident – Purchases of domestic assets by foreigners
- The website sold to Gabrielle is an export for the U.S., so
- U.S. exports and net exports increase.

EXAMPLE 1: Solutions

A. Kiara keeps the 5,000 euros at home

Kiara (a domestic resident) acquired a foreign asset (5,000 euros from Germany), so U.S. ***NCO*** increases

B. Kiara buys 5,000 euros worth of stocks in a German company

Kiara (a domestic resident) acquired a foreign asset (5,000 euros worth of stocks in a German company), so U.S. ***NCO*** increases

C. Kiara spends the 5,000 euros on German shoes

Kiara purchase of shoes made in Germany is an import for the U.S.: overall, exports increase by 5,000 euros (website sale to Germany), and imports increase by 5,000 euros (shoes bought from Germany), so U.S. Net exports do not change; NCO do not change.

D. Kiara exchanges the 5,000 euros into U.S. dollars

Now the bank has to use the 5,000 euros (keep in the vault, or purchase German assets, or sell the euros to an American that wants to purchase a good or service for euros) The change in NX is matched by the change in NCO.

EXAMPLE 2: Importing from China

Amazon U.S. purchases $10,000,000 worth of goods from China (to sell to the American customers). What is the effect on the U.S. net exports and net capital outflows if:

A. China buys $10 million worth of U.S. government bonds

B. China buys $10 million worth of goods from the U.S.

C. China buys $10 million worth of stocks in U.S. companies

Solutions:

The $10 million worth of goods bought from China are imports for the U.S. Imports increase, ***NX*** decrease.

A. and C. China buys $10 million worth of U.S. government bonds or stocks in U.S. companies

Purchase of domestic assets by foreigners increase, so U.S, ***NCO*** decrease

B. China buys $10 million worth of goods from U.S.

U.S. imports increase (Amazon buys goods from China) and U.S. exports increase (U.S. sells goods to China). ***NX*** do not change. ***NCO*** do not change.

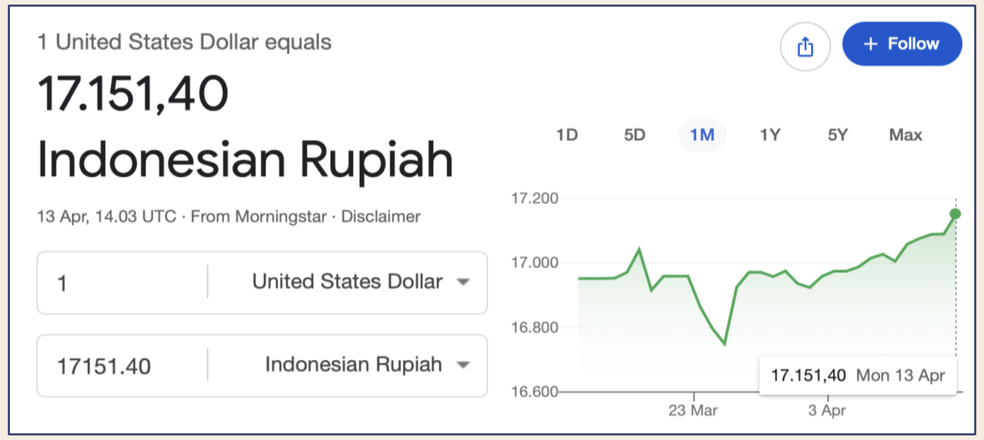
Nominal and Real Exchange Rate

The Nominal Exchange Rate

Nominal exchange rate:

Rate at which a person can trade the currency of one country for the currency of another

In domestic currency (IDR)	In foreign currency (USD)
USD/IDR	**IDR/USD**



IDR Terdepresiasi
USD Terapresiasi

Apresiasi

Penguatan nilai dari suatu mata uang, dihitung dengan jumlah mata uang lain yang bisa dibeli.

Depresiasi

Pelemahan nilai dari suatu mata uang, dihitung dengan jumlah mata uang lain yang bisa dibeli.

EXAMPLE 3: Appreciation or depreciation?

What is the exchange rate? Did the US dollar appreciated or depreciated?

A. Last year, Khalid exchanged $1 for 100 Japanese yen, but this year he exchanged $1 for 105 Japanese yen.

B. Last year, Amira exchanged $1 for 25 Mexican pesos, but this year she exchanged $1 for 20 Mexican pesos.

Solutions:

A. Last year's exchange rate = 100 yen per dollar

This year's exchange rate = 105 yen per dollar

US dollar appreciation : $1 now can buy more yens than last year (Yen depreciation)

B. Last year's exchange rate = 25 pesos per dollar

This year's exchange rate = 20 pesos per dollar

US dollar depreciation : $1 now can buy fewer pesos than last year (Peso appreciation)

Real Exchange Rate

*rate at which a person can trade the goods and services of one country for the **goods and services** of another.*

$$\text{Real exchange rate} = \frac{\text{Nominal exchange rate} \times \text{Domestic price}}{\text{Foreign price}}.$$

Understanding **RER** is important because it is the key determinant of how much a country imports/exports.

Example

Harga beras di Indonesia = **Rp 10.000** / kg
Harga beras di AS = **$4** / Kg

Dengan base currency USD, Maka,

RER = [(IDR/USD)*$4]/10.000
= [(17.151,40)*4]/10.000

6,86 KG Beras Indo per KG beras AS

For the Economy as a Whole: P = price of a domestic basket of goods relative to price of a foreign basket of goods

Apresiasi

Apresiasi dalam US RER berarti barang US akan semakin **mahal** → Mengurangi konsumsi dari barang AS dan meningkatkan konsumsi barang asing → Ekspor turun dan impor naik → NX dari US akan menurun.

Depresiasi

Depresiasi dari US RER berarti barang US akan semakin **murah** → Meningkatan konsumsi dari barang AS dan mengurangi konsumsi barang asing → Ekspor naik dan impor turun → NX dari US akan meningkat.

EXAMPLE 4: Calculating real exchange rate

A Big Mac costs $4 in U.S., 380 yen in Japan. The exchange rate is 110 yen per dollar. Compute the real exchange rate.

$\boldsymbol{e} \times \boldsymbol{P}$ = price in yen of a U.S. Big Mac

= (110 yen per $) x ($4 per Big Mac)

= 440 yen per U.S. Big Mac

Real exchange rate = $\boldsymbol{e} \times \boldsymbol{P} / \boldsymbol{P}^*$

= 440 yen per U.S. Big Mac / 380 yen per Japanese Big Mac

= 1.16 Japanese Big Macs per U.S. Big Mac

Active Learning 2: Compute a real exchange rate

The exchange rate is **e** = 20 pesos per $

The price of a tall Starbucks Latte is:

P = $3 in U.S. and ***P**** = 40 pesos in Mexico

A. What is the price of a U.S. latte measured in pesos?

B. Calculate the real exchange rate, measured as Mexican lattes per U.S. latte.

Answers:

e = 20 pesos per $; ***P*** = $3 in U.S., ***P**** = 40 pesos in Mexico

A. What is the price of a U.S. latte measured in pesos?

e x ***P*** = (20 pesos per $) x (3 $ per U.S. latte)

= 60 pesos per U.S. latte

B. Calculate the real exchange rate, measured as Mexican lattes per U.S. latte.

e x ***P*** / ***P**** = 60 pesos per U.S. latte / 40 pesos per Mexican latte = 1.5 Mexican lattes per U.S. latte

Purchasing Power Parity

Purchasing-Power Parity

- Purchasing-power parity
  - A theory of exchange rates, that says a unit of any given currency should be able to buy the same quantity of goods in all countries
- Based on the *law of one price*:
  - A good should sell for the same price in all locations
- Arbitrage
  - Take advantage of price differences for the same item in different markets

- PPP: a currency must have the same purchasing power in all countries
  - A U.S. dollar must buy the same quantity of goods in the United States and Japan
  - And a Japanese yen must buy the same quantity of goods in Japan and the United States
  - A unit of a currency must have the same real value in every country.

Purchasing Power Parity

Purchasing Power → **Nilai dari uang**

Parity → **Equality**

Based on the *law of one price*:

“ a good must sell for the same price in all locations.”

Explanation

Jika terdapat perbedaan harga, maka akan terjadi ***arbitrage***. Artinya, kita dapat beli barang di tempat lain dengan harga yang lebih murah, lalu menjual di tempat lain dengan harga yang lebih mahal.

Apabila hal ini terjadi dalam perdagangan internasional: harga 1kg kopi di AS lebih murah dibandingkan di Jepang, maka akan ada kelompok masyarakat yang membeli kopi di AS dan menjualnya di Jepang.

Maka, ekspor kopi dari AS yang meningkat akan meningkatkan harga di AS dan akan menurunkan harga di Jepang.

Implications of Purchasing Power Parity

Real exchange rate $= (e \times P)/P^*$.

$1/P = e/P^*$. → $1 = eP/P^*$. → Jika PPP berlaku, maka nilai tukar riil = 1 (harga sama di semua negara)

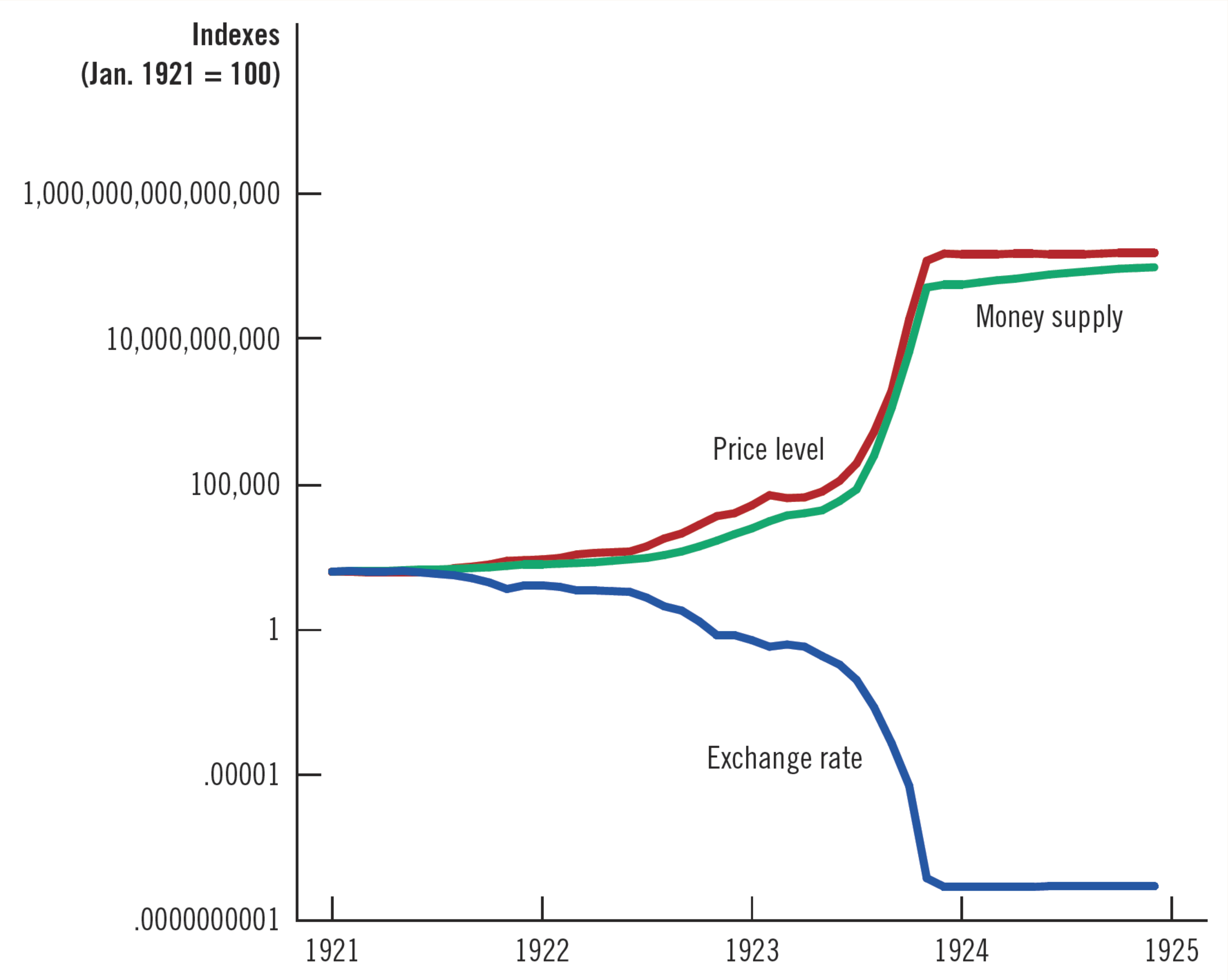
Jika daya beli dolar selalu sama di dalam negeri maupun di luar negeri, maka nilai tukar riil, yaitu harga relatif barang domestik dan barang luar negeri, tidak dapat berubah.

$e = P^*/P$.

Nilai tukar nominal antara mata uang dua negara harus mencerminkan tingkat harga di negara-negara tersebut. Nilai tukar nominal akan berubah ketika tingkat harga berubah.

→ When the central bank prints large quantities of money, that money loses value both in terms of the goods and services it can buy and in terms of the amount of other currencies it can buy

Hyperinflation in Germany, 1921-1924



Limitations

PPP is not completely accurate, as exchange rates changes don’t always move to ensure a dollar has the same real value in all countries all the time.

- Many goods are not easily traded.
  Example → if haircuts are cheaper in Jakarta than in Singapore, would people travel from Singapore to Jakarta just to get a haircut?

- Even tradable goods are not always the perfect substitute when they are produced in different countries.
  Example → If consumers in the US prefers BMW (German car) over Chevrolet (American car), and thus make prices of BMW rise, it won’t make them entirely shift from BMW to Chevrolet as the two goods are not perfect substitutes.

Thus, purchasing power parity is not the perfect theory for exchange rate determination.

EXAMPLE 5: The law of one price

If coffee beans sell for \$4/pound in Seattle and \$5/pound in Boston; and coffee beans can be costlessly transported, how will the two markets reach equilibrium?

- Opportunity for arbitrage: making a quick profit by buying coffee in Seattle and selling it in Boston.
- Seattle: increase demand drives up the price
- Boston: increase supply drives the price down
- Until the two prices are equal.

EXAMPLE 6: Applying PPP

Using the theory of purchasing-power parity, find the exchange rate if a bushel of apples sells for $30 in the U.S. and 300 krona in Sweden.

- PPP: a dollar buys the same quantity of goods in the United States (prices in $) as in Sweden (prices in krona)
- The number of krona per dollar must reflect the prices of goods in the U.S. and Sweden.
- Nominal exchange rate = 300 krona / 30 dollars = 10 krona per dollar

Active Learning 3: Chapter review questions

A. Which of the following statements about a country with a trade deficit is not true?

a) Exports < imports

b) Net capital outflow < 0

c) Investment < saving

d) $Y < C + I + G$

B. A Ford Escape Titanium Hybrid sells for $42,000 in the U.S. and 345,000 krona in Sweden. If purchasing-power parity holds, what is the nominal exchange rate (krona per dollar)?

Active Learning 3: Answers

A. Which of the following statements about a country with a **trade deficit** is <u>not true</u>?

a) Exports < imports

b) Net capital outflow < 0

c) Investment < saving

d) Y < C + I + G

- A trade deficit means: ***NX*** < 0; exports < imports;
- ***Y*** < domestic spending (***C*** + ***I*** + ***G***);
- and ***NCO*** < 0.
- Since ***NX*** = ***S*** – ***I***, a trade deficit implies ***I*** > ***S***.

B. A Ford Escape SUV sells for $42,000 in the U.S. and 345,000 krona in Sweden. If purchasing-power parity holds, what is the nominal exchange rate (rubles per dollar)?

- ***P**** = 345,000 krona
- ***P*** = $42,000
- ***e*** = ***P****/***P*** = 345000/42000 = 8.21 krona per dollar

THINK-PAIR-SHARE

You are watching a national news broadcast with your parents. The news anchor explains that the exchange rate for the dollar just hit its lowest value in a decade. The on-the-spot report shifts to a spokesman for Caterpillar, a heavy equipment manufacturer. The spokesman reports that sales of its earthmoving equipment have hit an all-time high and so has the value of its stock. Your parents are shocked by the report's positive view of the low value of the dollar. They just cancelled their European vacation because of the dollar's low value.

A. Why do Caterpillar and your parents have different opinions about the value of the dollar?
B. Caterpillar imports many parts and raw materials for their manufacturing processes, and they sell many finished products abroad. Since they are happy about a low dollar, what must be true about the proportions of their imports and exports?
C. If someone argues that a strong dollar is "good for America" because Americans are able to exchange some of their GDP for a greater amount of foreign GDP, is it true that a strong dollar is good for every American? Why or why not?

1. Pada awal Juni 2021, McDonald's merilis paket makanan baru yang terinspirasi dari *boyband* populer dari Korea Selatan, BTS. Paket makanan tersebut bernama ‘McD BTS Meal’, terdiri dari 10 potong McNugget, kentang goreng sedang, Coke ukuran sedang, serta saus cabai manis dan Cajun. Jika harga paket McD BTS di Singapura adalah $5 dan harga McD BTS di Malaysia adalah RM14, dan nilai tukarnya adalah 3RM per SGD, berapakah nilai tukar riil (*real exchange rate*)?
   a. 4/15 *BTS Meal* Malaysia per *BTS Meal* Singapura.
   b. 15/14 *BTS Meal* Malaysia per *BTS Meal* Singapura.
   c. 42/5 *BTS Meal* Malaysia per *BTS Meal* Singapura.
   d. 5/42 *BTS Meal* Malaysia per *BTS Meal* Singapura.

2. Pemerintah negara X berkomitmen memerangi korupsi dan memberikan pembebasan pajak bagi investor untuk memperbaiki iklim bisnis dan mengurangi risiko aset. Dampak dari kebijakan tersebut bagi negara X adalah ….
   a. meningkatnya ekspor neto dan *net capital outflow*
   b. meningkatnya ekspor neto dan mengurangi *net capital outflow*
   c. menurunnya ekspor neto dan *net capital outflow*
   d. menurunnya ekspor neto dan meningkatnya *net capital outflow*

3. Perhatikan data konversi *Purchasing Power Parity* (PPP) pada tabel berikut. Data menunjukkan daya beli ekuivalen dari 1 USD pada tiga negara di tahun 2021.

Tabel 1. Konversi PPP, 2021 (mata uang domestik per USD) 2021

Negara	**Kurs**
Indonesia (IDR)	4.738,64
Jepang (JPY)	61,47
Taiwan, ROC (TWD)	13,48

Sumber: OECD,2022

Jika harga roti lapis isi tuna ‘Subway’ di Indonesia adalah Rp37.500,00, berapakah harganya di Taiwan? (Petunjuk: nilai konversi PPP adalah nilai tukar riil terhadap USD)

a. TWD342,39.
b. TWD109,53.
c. TWD2.709,54.
d. TWD7,91.

4. Biaya berlangganan paket Netflix standar di Taiwan adalah **TWD330** dan Rp153.000,00 di Indonesia. Diketahui nilai tukar nominal yang berlaku saat ini adalah Rp494,73 per TWD. Perhatikan pernyataan-pernyataan berikut.
   (1) Nilai tukar kedua negara berdasarkan PPP adalah sebesar Rp463,64 per TWD.
   (2) Berdasarkan PPP, nilai tukar rupiah saat ini 6,28% lebih rendah *(undervalued)*.
   (3) Biaya langganan dua paket standar Netflix di Indonesia ekuivalen dengan TWD618,52.
   (4) Biaya berlangganan paket Netflix di Taiwan lebih murah daripada di Indonesia.

   Manakah dari keempat pernyataan tersebut yang benar?
   a. (1), (2), dan (3).
   b. (1) dan (3).
   c. (2) dan (4).
   d. (4).